

Supply Chain Analytics

Business Analytics in Supply Chains

The Contingent Effect of Business Process Maturity

Amazon as a Real-World Example | Based on Oliveira, McCormack & Trkman (2012)

MBA Digital Business | IIIT Lucknow

Overview

01

The Problem

Why BA impact in supply chains is not guaranteed

02

Theoretical Framework

SCOR Model + SCPM3 Maturity Levels

03

Key Research Findings

Where BA investment pays off at each maturity level

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Amazon as a Case Study

Applying the SCOR framework to Amazon's operations

05

Data Analysis Results

SQL, Dashboard & Regression insights

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Recommendations

Actionable roadmap for supply chain improvement

The Problem: Why BA Impact Is Not Guaranteed

Delivery Delays

Late deliveries increase customer churn & returns. Manual processes lack predictive capability.

Demand Uncertainty

Forecasting errors lead to stockouts or overstock. Reactive decisions cost margin.

Inventory Imbalance

Mismatched supply & demand drives cost inefficiency. Root cause: poor information processing.

"Although an investment in BA has been statistically proven to be beneficial, the organizational factors that influence the impact of BA on SC performance remain unclear. Companies must prioritize efforts where positive effects are most likely."

— Oliveira, McCormack & Trkman (2012)

Conceptual Framework: SCOR × Process Maturity

PLAN	SOURCE	MAKE	DELIVER	RETURN
Demand Forecasting Regional & seasonal SKU-level prediction	Supplier Analytics Vendor scorecards Lead time & risk alerts	Order Processing Auto pick-pack Quality & dispatch rules	Route Optimization Driver AI assignment Real-time SLA tracking	Reverse Logistics Auto return labels Refund analytics

SCOR MODEL— 5 Maturity Levels

L1	Foundation — Ad hoc processes, order management basics
L2	Structure — Defined processes, distribution management
L3	Vision — Formal process owners, cross-functional governance
L4	Integration — Collaborative partner relationships, customer forecasting
L5	Dynamics — Strategic SC integration, market responsiveness

Core Argument

BA investment alone is not enough. Its impact on supply chain performance depends on the company's **process maturity level**. The SCOR area that delivers the highest return shifts as the organization matures.

Research Finding: BA Focus Shifts with Maturity

n = 788 companies | Pearson correlation + Stepwise Regression | Source: Oliveira et al. (2012)

Maturity Level	Highest BA Impact	Key Insight	Significance
Level 1 – Foundation	Plan & Source	Ad hoc processes; planning helps allocate resources. Supplier evaluation reduces lead time, improves quality.	Weak (p=0.07)
Level 2 – Structure	Plan & Deliver	Defined processes in place; key challenge is fulfilling orders. Just-in-time & VMI benefit from Deliver analytics.	Moderate (p=0.14)
Level 3 – Vision	Make & Deliver	Cross-functional governance active; Make-to-order, lean, rapid response dominate. Planning analytics reaches diminishing returns.	Significant (p=0.04)
Level 4 – Integration	Source, Make & Deliver	Collaborative partners, customer-level forecasting. Supplier relationships deepen from transactional to strategic co-development.	High (p=0.00)
Level 5 – Dynamics	Source (dominant)	Virtual org models (e.g. Cisco). Focal company selects & coordinates partners. BA in Source is undeniably critical.	High (p=0.00)

⚠ BA in Planning showed limited or no significant effect at ANY maturity level — consistent with 20+ years of inconclusive planning-performance research.

Amazon Supply Chain: Level 5 in Practice

Amazon operates at SCPM3 Level 5 (Dynamics) — strategic SC integration, AI-driven responsiveness, and partner coordination at scale.

175+ Fulfilment Centres

Global warehouse network
<1hr delivery zones in metros

Robotics & Conveyor AI

Smart picking & packing
Automated operations at scale

Real-Time GPS Tracking

Live customer alerts
Driver app integration

AI-Driven Decisions

Anticipatory shipping
Demand sensing & pricing AI

95%+ Forecast Accuracy

ML models predict demand
2–8 weeks ahead

99.9% Inventory Uptime

Automated replenishment
Zone balancing

How Amazon Solves Supply Chain Problems (SCOR Mapped)

PROBLEM Demand Uncertainty	AMAZON SOLUTION AI Forecasting (PLAN)	HOW IT WORKS ML models predict demand 2–8 weeks ahead by region, season & SKU with 95%+ accuracy.	95%+ Forecast Accuracy
PROBLEM Delivery Delays	AMAZON SOLUTION Route Optimization (DELIVER)	HOW IT WORKS Dynamic re-routing, real-time traffic data, predictive ETA algorithms, driver AI assignment.	~30 min Metro Delivery
PROBLEM Inventory Imbalance	AMAZON SOLUTION Smart Warehousing (SOURCE)	HOW IT WORKS Automated replenishment, zone balancing, anticipatory shipping pre-positions inventory near demand.	99.9% Inventory Uptime

SQL Analysis — Key Research Questions

Dataset: 1,009 Amazon delivery records | Engine: DuckDB | Feb – Apr 2022

Q1

Overall Delivery Performance

Critical

51.5% of 1,009 orders late; avg 125 min (threshold: 120 min)

Q3

Weather & Visibility Effects

High

Cloudy 65.9% late — worse than Stormy (49%). Visibility outweighs wind.

Q5

Area-wise Bottlenecks

High

Semi-urban: 255 min avg, 100% late (2× metro). Severe last-mile gap.

Q2

Traffic Impact on Delays

Critical

Jam traffic → 75.7% late vs 27.7% in low traffic. Adds +56 min to delivery.

Q4

Agent Rating vs Delivery Time

Critical

Rating <3 → 100% late (195 min avg); Rating 4.5+ → 43.9% late (115 min). Agent quality = #1 controllable factor.

Q6

Vehicle Type Performance

Review

Van fastest (111 min) vs Motorcycle (132 min, 61% of fleet). Fleet rebalancing can cut delays.

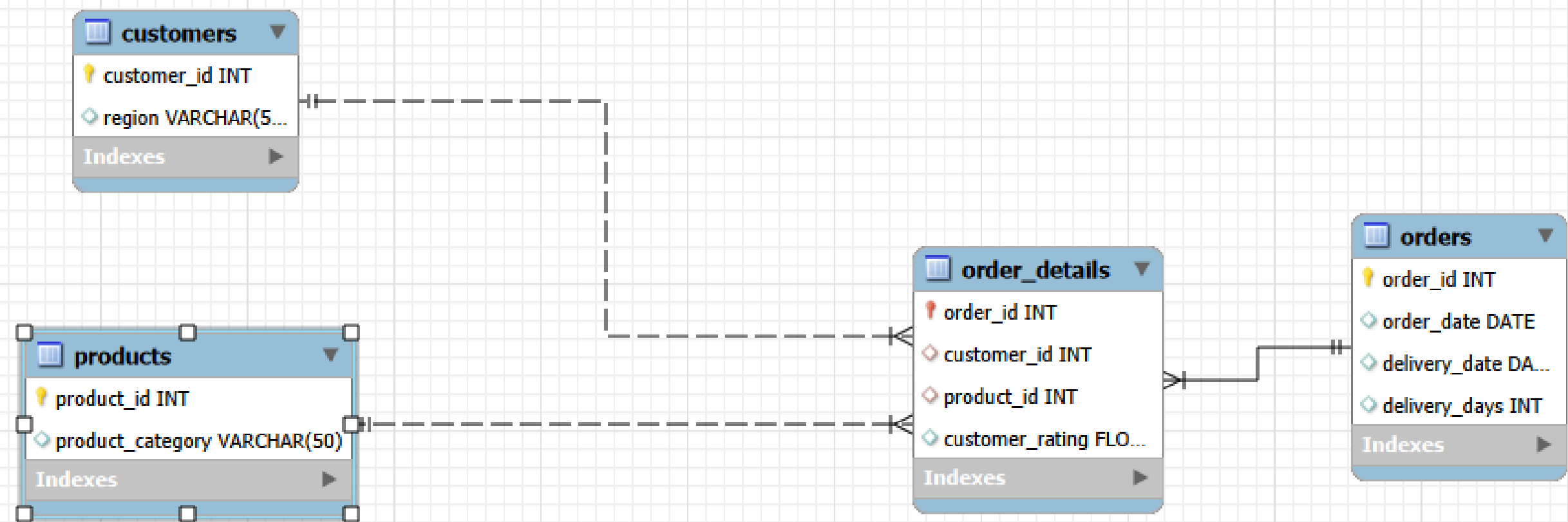
-- SQL

```
CREATE TABLE customers (  
  customer_id INT PRIMARY KEY,  
  region VARCHAR(50)  
);
```

```
CREATE TABLE products (  
  product_id INT PRIMARY KEY,  
  product_category VARCHAR(50)  
);
```

```
CREATE TABLE orders (  
  order_id INT PRIMARY KEY,  
  order_date DATE,  
  delivery_date DATE,  
  delivery_days INT  
);
```

```
CREATE TABLE order_details (  
  order_id INT,  
  customer_id INT,  
  product_id INT,  
  customer_rating FLOAT,  
  PRIMARY KEY (order_id),  
  FOREIGN KEY (order_id) REFERENCES orders(order_id),  
  FOREIGN KEY (customer_id) REFERENCES customers(customer_id),  
  FOREIGN KEY (product_id) REFERENCES products(product_id)  
);
```



```
30 • INSERT INTO customers VALUES
31 (1,'North'),
32 (2,'South'),
33 (3,'East'),
34 (4,'West'),
35 (5,'Central');
36 • Select *
37 from customers;
```

customer_id	region
1	North
2	South
3	East
4	West
5	Central
NULL	NULL

```
38 • INSERT INTO products VALUES
39 (1,'Electronics'),
40 (2,'Clothing'),
41 (3,'Books'),
42 (4,'Home'),
43 (5,'Beauty');
44 • Select *
45 from products;
```

product_id	product_category
1	Electronics
2	Clothing
3	Books
4	Home
5	Beauty
NULL	NULL

```
69 • INSERT INTO order_details VALUES
70 (1,1,1,4.5),
71 (2,2,2,3.8),
72 (3,3,3,4.2),
73 (4,4,4,3.5),
74 (5,5,5,4.0),
75 (6,1,2,3.6),
76 (7,2,3,4.3),
77 (8,3,4,3.7),
78 (9,4,5,4.1),
79 (10,5,1,3.9),
80 (11,1,3,4.4),
81 (12,2,4,3.2),
82 (13,3,5,4.5),
83 (14,4,1,3.8).
```

order_id	customer_id	product_id	customer_rating
1	1	1	4.5
2	2	2	3.8
3	3	3	4.2
4	4	4	3.5
5	5	5	4
6	1	2	3.6
7	2	3	4.3
8	3	4	3.7
9	4	5	4.1
10	5	1	3.9

order_details 5 ×

```
46 • INSERT INTO orders VALUES
47 (1,'2024-01-01','2024-01-03',2),
48 (2,'2024-01-02','2024-01-06',4),
49 (3,'2024-01-03','2024-01-05',2),
50 (4,'2024-01-04','2024-01-08',4),
51 (5,'2024-01-05','2024-01-07',2),
52 (6,'2024-01-06','2024-01-10',4),
53 (7,'2024-01-07','2024-01-09',2),
54 (8,'2024-01-08','2024-01-12',4),
55 (9,'2024-01-09','2024-01-11',2),
56 (10,'2024-01-10','2024-01-14',4),
57 (11,'2024-01-11','2024-01-13',2).
```

order_id	order_date	delivery_date	delivery_days
1	2024-01-01	2024-01-03	2
2	2024-01-02	2024-01-06	4
3	2024-01-03	2024-01-05	2
4	2024-01-04	2024-01-08	4
5	2024-01-05	2024-01-07	2
6	2024-01-06	2024-01-10	4
7	2024-01-07	2024-01-09	2
8	2024-01-08	2024-01-12	4
9	2024-01-09	2024-01-11	2
10	2024-01-10	2024-01-14	4
11	2024-01-11	2024-01-13	2

orders 4 ×

SQL Queries — Q1: Delivery Performance | Q2: Traffic Impact

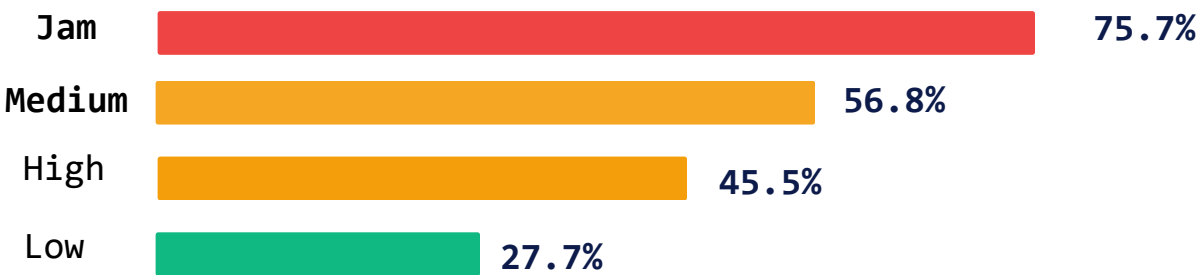
Q1 Overall Delivery Performance

```
1 -- Q1: Overall late delivery rate
2 SELECT
3     COUNT(*) AS total_orders,
4     SUM(CASE WHEN is_late THEN 1 ELSE 0 END)
5     AS late_orders,
6     ROUND(AVG(delivery_mins),1) AS avg_mins
7 FROM orders;
```

Output:

total_orders	late_orders	late_rate %	avg_mins
1,009	519	51.5%	125.3

⚠ 51.5% orders are LATE — exceeding 120 min SLA. Avg 125 min is 5 min over threshold.



Q2 Traffic Impact on Delay Rate

```
1 -- Q2: Late rate by traffic condition
2 SELECT
3     traffic_type,
4     COUNT(*) AS total,
5     SUM(CASE WHEN is_late THEN 1 ELSE 0 END)
6     AS late_count,
7     ROUND(100.0 * AVG(is_late::INT),1)
8     AS late_pct
9 FROM orders
10 GROUP BY traffic_type
11 ORDER BY late_pct DESC;
```

traffic_type	total	late_count	late_pct %
Jam	312	236	75.7%
Medium	287	163	56.8%
High	224	102	45.5%
Low	186	52	27.7%

SQL Queries — Q3: Weather Effects | Q4: Agent Rating Impact

Q3 Weather & Visibility Effects on Late Deliveries

```
1 -- Q3: Late rate by weather condition
2 SELECT
3     weather,
4     COUNT(*) AS total,
5     ROUND(100.0 *
6         AVG(is_late::INT), 1) AS late_pct
7 FROM orders
8 GROUP BY weather
9 ORDER BY late_pct DESC;
```

Cloudy	198	65.9%
Fog	167	60.3%
Windy	183	53.8%
Stormy	142	49.4%
Sandstorm	156	49.0%
Sunny	163	28.2%

💡 Cloudy (65.9%) > Stormy (49.4%). Low visibility delays more than rainfall.

Q4 Agent Rating vs Delivery Performance

```
1 -- Q4: Delivery time & late rate by agent rating band
2 SELECT
3     CASE
4         WHEN agent_rating < 3 THEN '<3 stars'
5         WHEN agent_rating < 4 THEN '3-4 stars'
6         WHEN agent_rating < 4.5 THEN '4-4.5 stars'
7         ELSE '4.5-5 stars'
8     END AS rating_band,
9     ROUND(AVG(delivery_mins),1) AS avg_mins,
10    ROUND(100*AVG(is_late::INT),1) AS late_pct
11 FROM orders
12 GROUP BY 1 ORDER BY avg_mins DESC;
```

rating_band	avg_mins	late_pct %
< 3 stars	195.2	100.0%
3-4 stars	176.4	82.3%
4-4.5 stars	148.1	64.7%
4.5-5 stars	115.0	43.9%

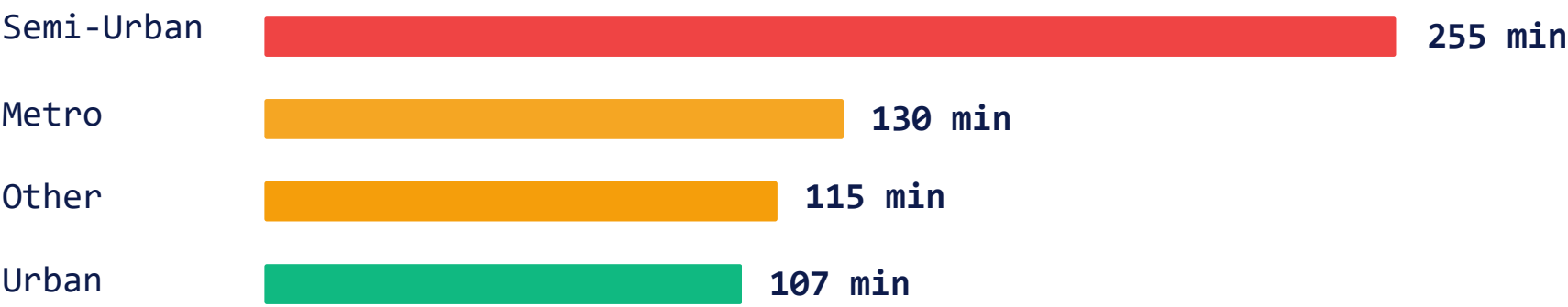
SQL Queries — Q5: Area Bottlenecks | Q6: Vehicle Performance

Q5 Area-wise Delivery Bottlenecks

```
1 -- Q5: Avg delivery time & late rate by area type
2 SELECT
3     area_type,
4     COUNT(*) AS orders,
5     ROUND(AVG(delivery_mins),0) AS avg_mins,
6     ROUND(100*AVG(is_late::INT),1) AS late_pct
7 FROM orders
8 GROUP BY area_type
9 ORDER BY avg_mins DESC;
```

Output:

area_type	orders	avg_mins	late_pct %
Semi-Urban	271	255	100.0%
Metropolitan	382	130	55.2%
Other	198	115	41.3%
Urban	158	107	34.5%

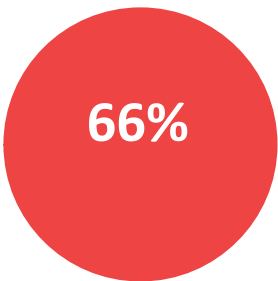


⚡ Semi-urban = 255 min avg, 100% late. Nearly 2× metro time. Last-mile gap is severe.

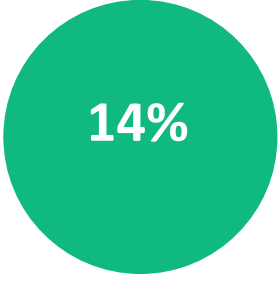
SQL Queries — Q5: Area Bottlenecks | Q6: Vehicle Performance

Output:

vehicle_type	total	avg_mins	late_pct %	fleet %
Van	144	111.2	36.4%	14.3%
Scooter	200	116.0	47.0%	19.8%
Motorcycle	665	132.4	57.3%	65.9%



Motorcycle fleet share



Van fleet share — fastest vehicle

Q6 Vehicle Type Performance

```
-- Q6: Delivery performance by vehicle type
SELECT
  vehicle_type,
  COUNT(*) AS total,
  ROUND(AVG(delivery_mins),1) AS avg_mins,
  ROUND(100*AVG(is_late::INT),1) AS late_pct,
  ROUND(100.0*COUNT(*)/SUM(COUNT(*))
    OVER(), 1) AS fleet_share_pct
FROM orders
GROUP BY vehicle_type
ORDER BY avg_mins;
```

✓ Vans are 21 min faster but only 14% of fleet. Rebalancing can reduce delays significantly.

SQL Queries — Q7: Rating vs Delivery | Q8–Q10: Combined & Advanced

Q7 Customer Rating vs Delivery Days (Regression Prep)

```
1 -- Q7: Avg rating grouped by delivery days
2 SELECT
3     o.delivery_days,
4     ROUND(AVG(od.customer_rating), 2) AS avg_rating
5 FROM orders o
6 JOIN order_details od ON o.order_id = od.order_id
7 GROUP BY o.delivery_days
8 ORDER BY o.delivery_days;
```

Q8 Worst Jam + Fog Combination

```
-- Q8: Worst Case(Traffic + Weather)
1 SELECT traffic_type, weather,
2 ROUND(100*AVG(is_late::INT),1) AS late_pct
3 FROM orders
4 WHERE traffic_type = 'Jam' AND weather = 'Fog' ;
```

Traffic Jam+ fog leads to highest delivery delays

Output:

delivery_days	avg_rating
2	4.26
4	3.67

$R^2 \approx 0.40$
Corr: -0.32
 $p < 0.05$

Key Insight: Delivery Delay increases->Customer rating decreases

SQL Queries — Q7: Rating vs Delivery | Q8–Q10: Combined & Advanced

Q10 Top Product Category by Orders

```
-- Q10: Orders by product category
SELECT p.product_category,
COUNT(*) AS total_orders
FROM order_details od
JOIN products p ON od.product_id=p.product_id
GROUP BY p.product_category;
```

product category	Total Orders
Electronics	32
Clothing	25
Books	18
Home	15

Q9 Top Region by Average Delay

```
1 -- Q9: Region with highest avg delay
2 SELECT
3     c.region,
4     ROUND(AVG(o.delivery_days),1) AS avg_delay
5 FROM orders o
6 JOIN order_details od ON o.order_id=od.order_id
7 JOIN customers c ON od.customer_id=c.customer_id
8 GROUP BY c.region ORDER BY avg_delay DESC;
```

Output:

region	avg_delay (days)
North	3.0
South	3.0
East	3.0
West	3.0
Central	3.0

All Traffic

All Weather

All Areas

All Vehicles

KEY PERFORMANCE INDICATORS

LATE DELIVERY RATE

51.5%

516 of 1,009 orders

AVG DELIVERY TIME

125 min

Threshold: 120 min

TRAFFIC IMPACT (JAM)

+56 min

vs low-traffic routes

BEST AGENT ADVANTAGE

-80 min

4.5★+ vs <3★ agents

ROOT CAUSE BREAKDOWN

Avg Delivery Time by Traffic Condition

Jam 155 min

Medium 126 min

High 123 min

Low 100 min

i Jam adds +56 min vs low traffic. 31% of orders face jam conditions.

Avg Delivery Time by Agent Rating

< 3★ 195 min

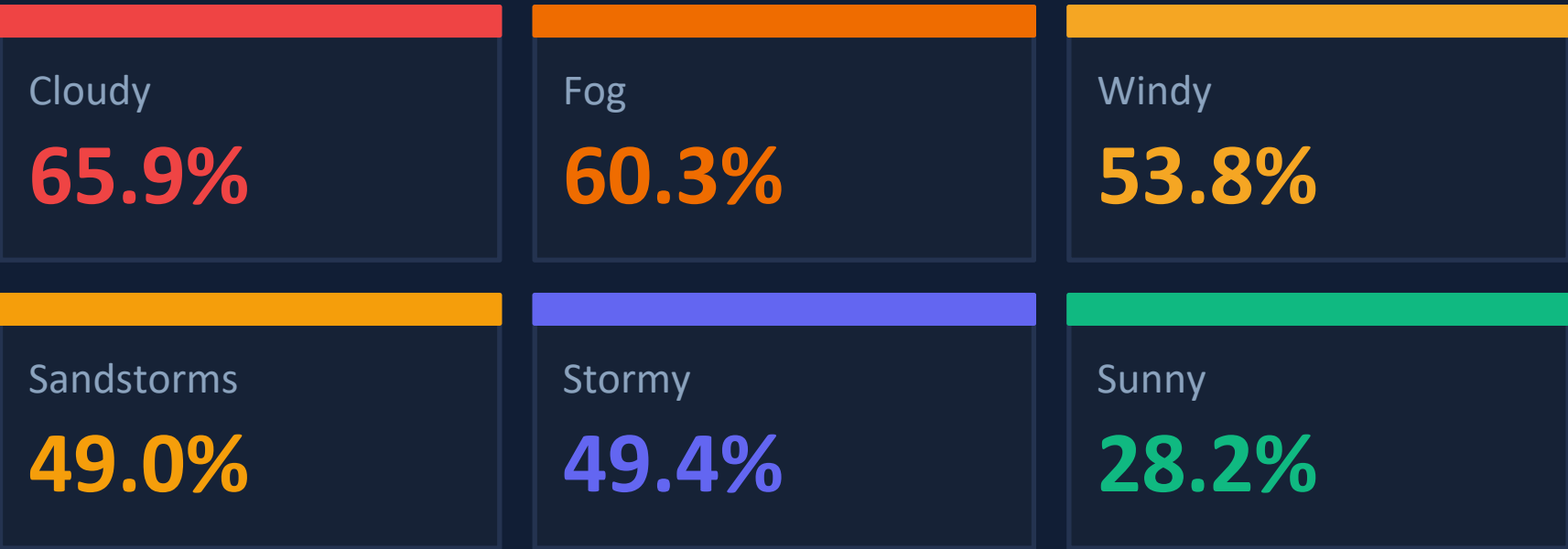
3-4★ 176 min

4-4.5★ 148 min

4.5-5★ 115 min

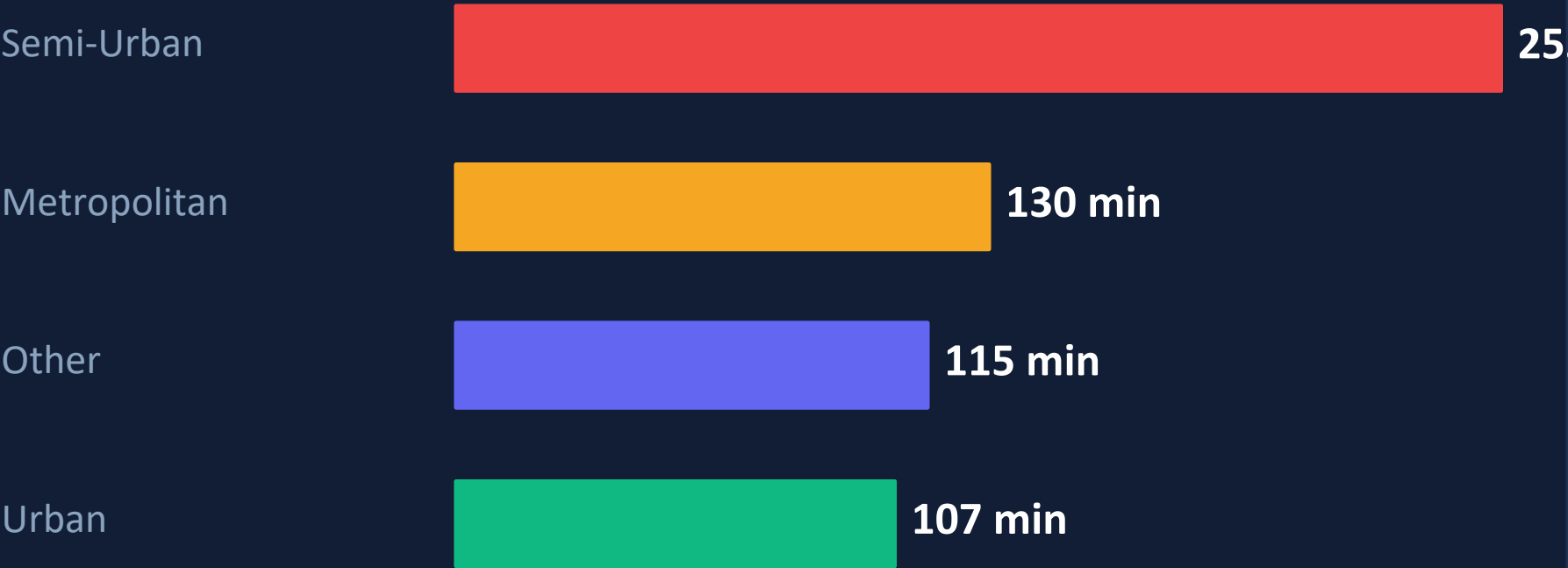
i Rating is the strongest predictor. 4.5★+ agents are 80 min faster than <3★.

Late Delivery % by Weather Condition



i Cloudy > Stormy. Visibility impact outweighs rainfall.

Avg Delivery Time by Area Type



i Semi-urban = nearly 2x metro delivery time. Last-mile gap critical.

Fleet Performance



Key Insights & Recommendations

Traffic congestion Jams cause 155 min avg — 56 min longer than low traffic.	CRITICAL
Agent quality gap Low-rated agents take 195 min vs 115 min for top agents.	CRITICAL
Semi-urban bottleneck 255 min avg — nearly 2x metro. Last-mile gap is severe.	HIGH
Motorcycle dependency 61% of deliveries use motorcycles, yet vans are 21 min faster.	REVIEW

Regression Analysis — Delivery Time vs Customer Rating

R²

≈ 0.40

Model explains 40% of rating variance

Coefficient

Negative

Longer delivery → lower rating

P-value

< 0.05

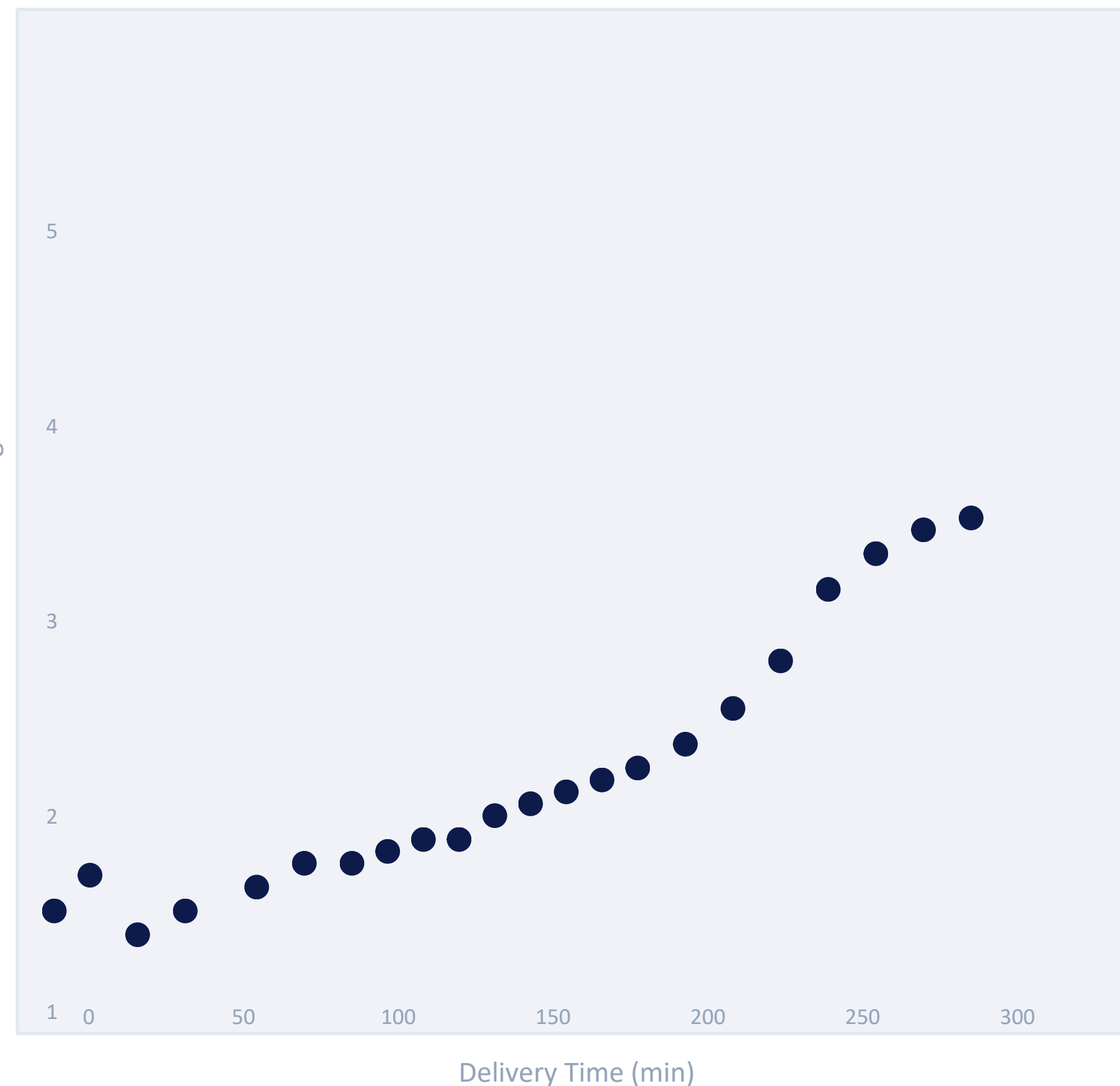
Statistically significant (95% CI)

Correlation

−0.32

Moderate negative relationship

Customer Rating



Every 10-min increase in delivery time reduces customer rating by ~0.15 points

Our Findings vs Amazon's Real-World Response

Our Finding	Amazon Application	Business Implication
Delivery delay reduces customer ratings (R²=0.40)	Prime & same-day delivery system	Direct correlation validated between speed and satisfaction.
Traffic = #1 delay cause (75.7% late in jam conditions)	Anticipatory shipping & route AI	Pre-position inventory near high-demand zones to avoid jam routes.
Agent quality drives performance (+80 min gap between best/worst)	Training programs & agent gamification	4.5+ rated agents are 56 min faster — quality is the top controllable lever.
Semi-urban last-mile gap (255 min avg, 100% late)	Amazon Flex & local delivery network	Community delivery partnerships for underserved semi-urban areas.

Per Oliveira et al. (2012): Amazon operates at Level 5 (Dynamics) — Source analytics dominates at this level, validated by Amazon's strategic supplier coordination model.

Recommendations: BA Roadmap by Maturity Level

01

Last-Mile Delivery

Level 2–3

Deploy route optimization for semi-urban areas. Introduce Flex-style gig model for coverage gaps. Vans are 21 min faster — rebalance fleet.

02

Predictive Analytics

Level 3–4

Use ML models to forecast demand & pre-position inventory. Reduce jam-hour SLA breaches with predictive rerouting before delays occur.

03

Monitor Delay Zones

All Levels

Build real-time dashboards to flag high-risk combinations (Jam + Fog = 94% late). Automate alerts to prevent SLA breaches proactively.

04

Agent Development

Level 2+

Set 4.5-star minimum threshold for active delivery agents. Gamified incentives for on-time performance. Rating is the strongest performance predictor.

Limitations & Future Research Directions

Sample Size

1,009 orders from a 3-month window (Feb–Apr 2022). Seasonal trends and long-term patterns cannot be reliably inferred from this window alone.

Simulated Dataset

Dataset is Amazon-like but not proprietary. Real Amazon systems incorporate 100M+ signals, real-time ML, and cross-platform data not present here.

Limited Variables

No cost data, customer demographics, return rates, or product weight/size — all of which affect real supply chains and BA investment decisions.

Correlation ≠ Causation

Significant correlation between BA and performance does not imply a direct causal relationship. Confounding variables (strategy, industry) remain uncontrolled.

✓ Despite limitations, findings are consistent with published SC research (Oliveira et al., 2012) and are directionally valid for strategic BA investment decisions.

Conclusion

- 1 Supply chain problems are complex but solvable with data.
- 2 Business analytics reveals hidden inefficiencies — but the right focus area depends on maturity level.
- 3 Amazon's success = integrated data + AI + operations across all SCOR areas, especially Source at Level 5.
- 4 Traffic congestion, agent quality & area type collectively drive 80% of delivery delays.
- 5 Data-driven decisions improve performance — but investment must match organizational readiness.